

CALCULUS III

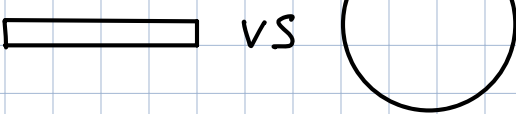
MATH 20550 , SECTION 04

Marianna Russkikh
mrusskik@nd.edu
126 Hayes-Healy

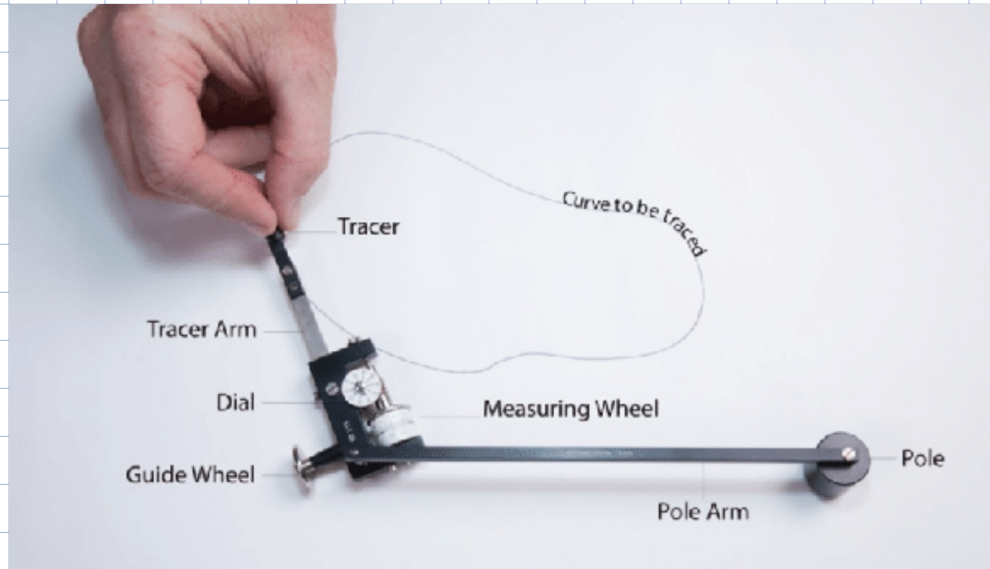
Motivation:

Polar Planimeter

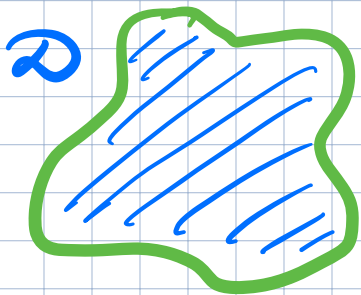
- measures the area by just going around the perimeter

Q:  vs

A: It takes in account how the angle changes when go around.



Green's Thm:

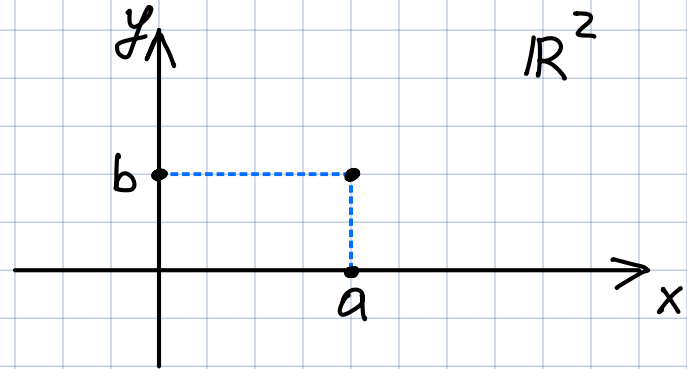


$$\oint_{\partial D} (L dx + M dy) = \iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dA$$

Three-dimensional coordinate system

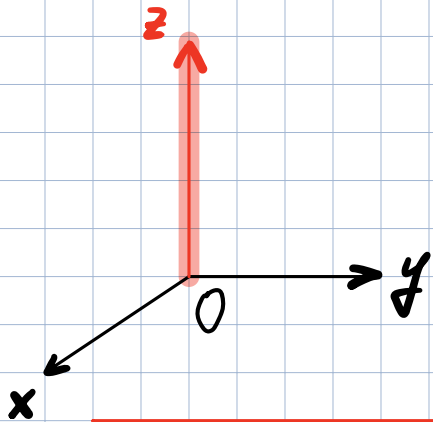
Reminder: A point on a plane is given by a pair of numbers

(a, b)
↑ ↑
x-coordinate y-coordinate



• A point in space ($\mathbb{R}^3$) is given by triple of numbers

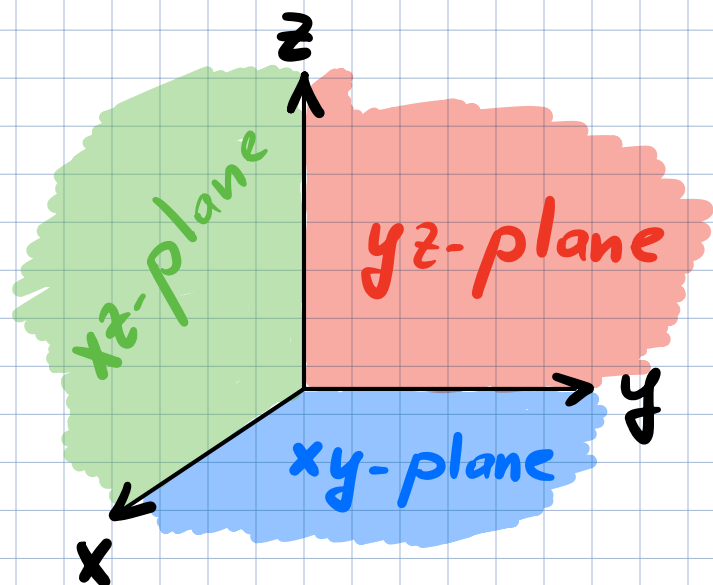
(a, b, c)
↑ ↑ ↑
x-coord. y-coord. z-coordinate



the direction of the z-axis is determined by the **right-hand-rule**

• Choose a fixed point O (the origin) and 3 directed, mutually perpendicular

lines through O :
x-axis
y-axis
z-axis

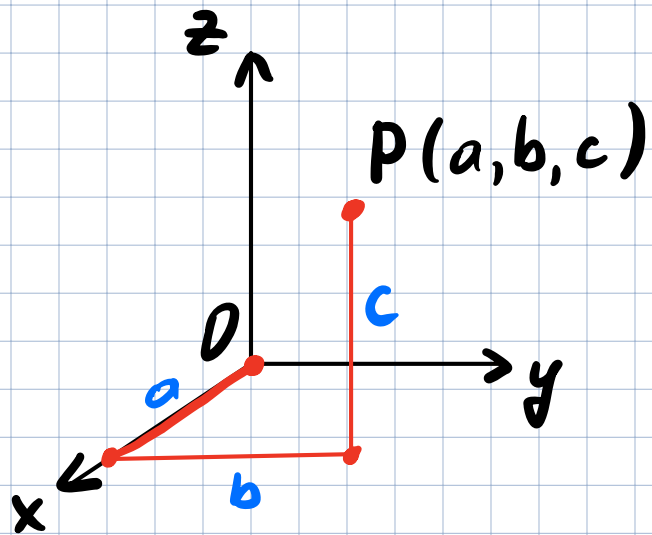


- Coordinate axes determine coordinate planes:

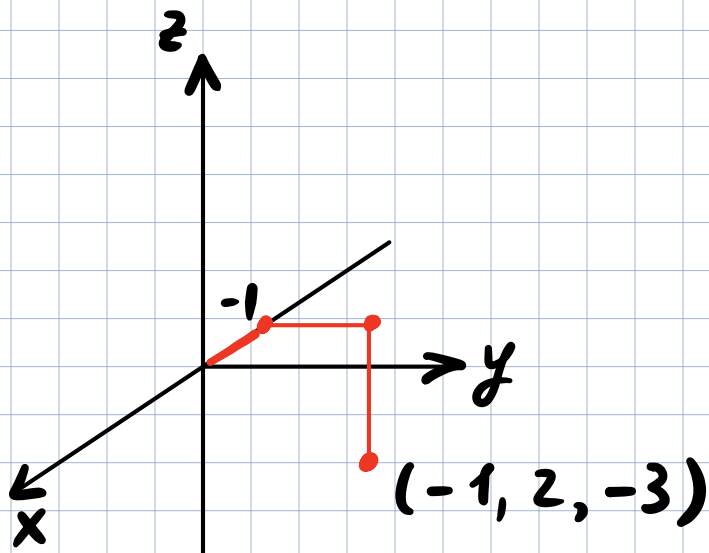
xy-plane
yz-plane
xz-plane

- Coordinate planes divide space into 8 octants

1st octant = all points (a, b, c)
with $a, b, c > 0$

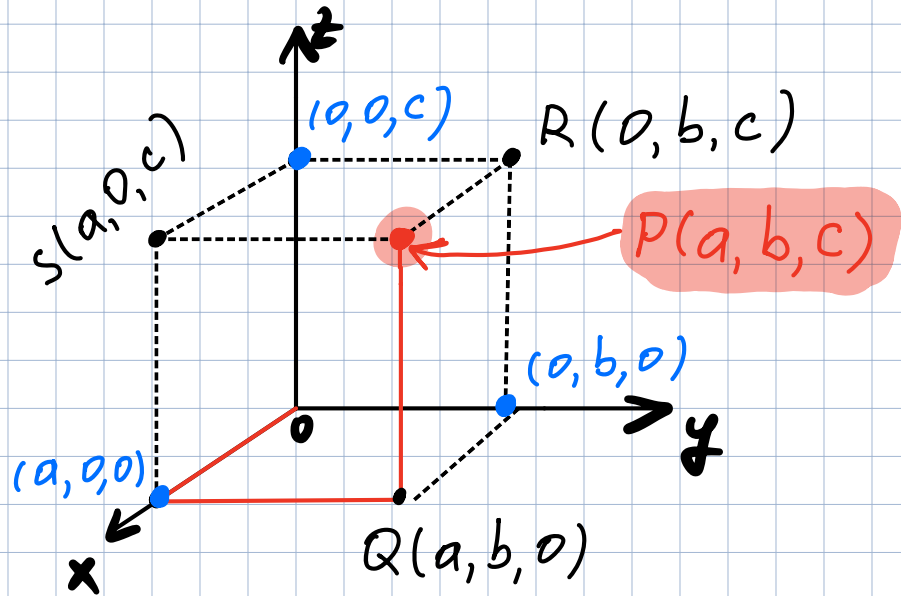


- a is the (directed) distance From P to the yz -plane
- b is the (directed) distance From P to the xz -plane
- c is the (directed) distance From P to the xy -plane



NB: coordinates can be negative!

The point $P(a, b, c)$ determines a rectangular box:



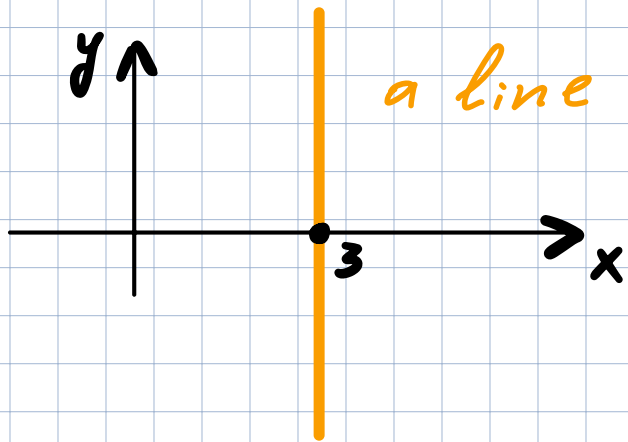
- $Q(a, b, 0)$ is the projection of P onto xy -plane
- $R(0, b, c)$ is the projection of P onto yz -plane
- $S(a, 0, c)$ is the projection of P onto xz -plane

Surfaces

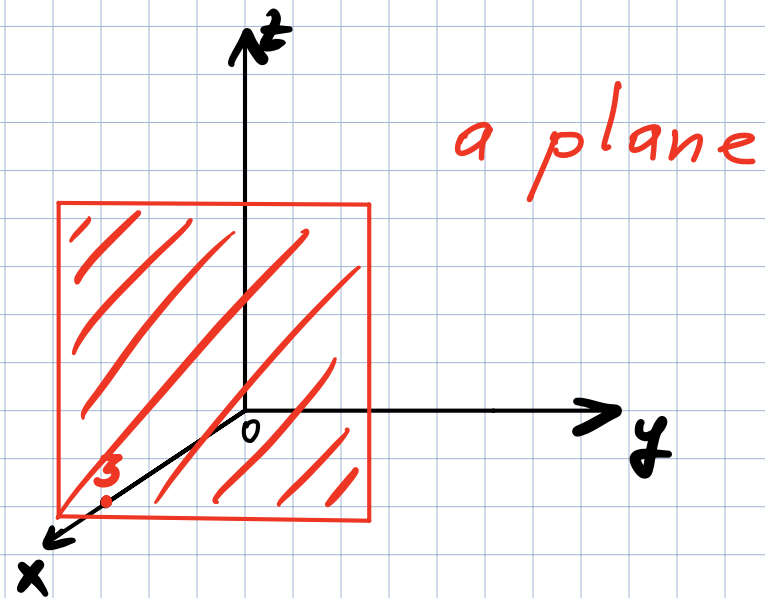
On $\mathbb{R}^2$, graph of an equation involving x, y represents a curve in $\mathbb{R}^2$

In $\mathbb{R}^3$, an equation involving x, y, z represents a surface in $\mathbb{R}^3$

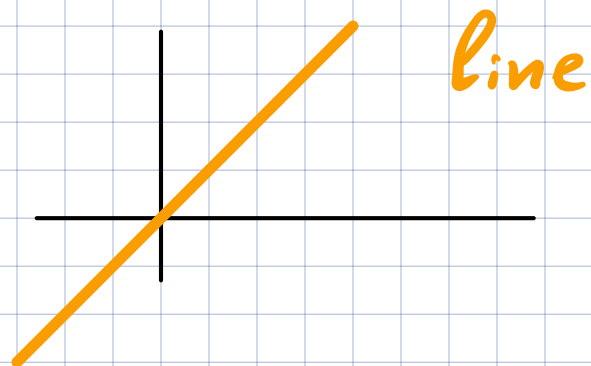
Ex: $\{(x, y) \mid x = 3\}$



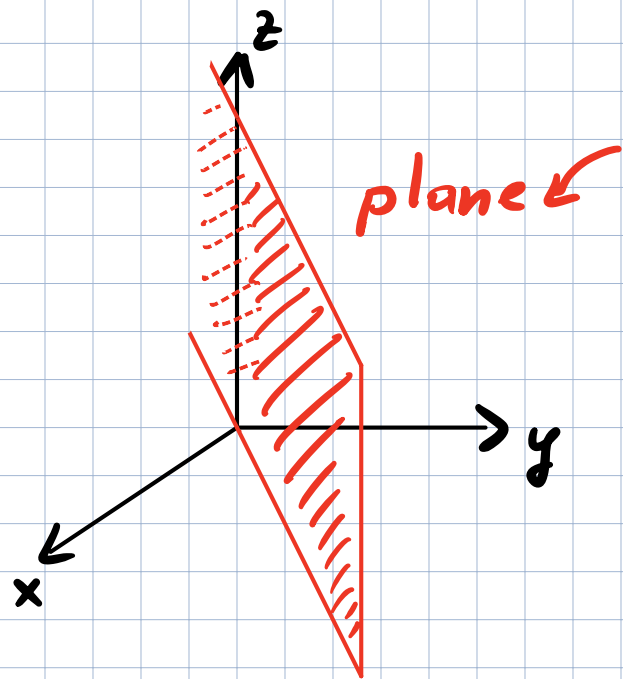
$\{(x, y, z) \mid x = 3\}$



Ex: In $\mathbb{R}^2$: $x=y$



In $\mathbb{R}^3$: $x=y$



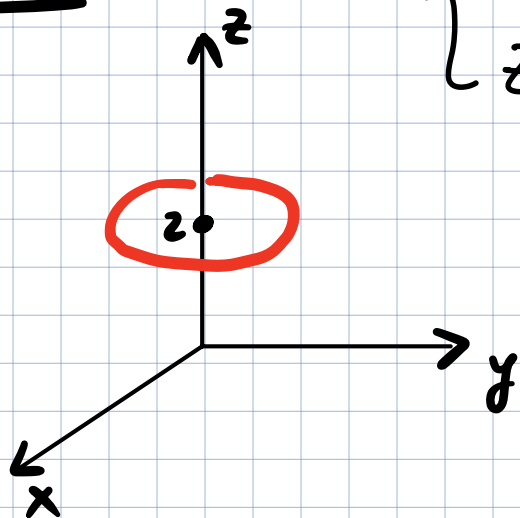
vertical (i.e. perpendicular
to xy -plane)
and intersecting it in
the line $\begin{cases} z=0 \\ x=y \end{cases}$

Ex:

(a)
$$\begin{cases} x^2 + y^2 = 1 \\ z = 2 \end{cases}$$

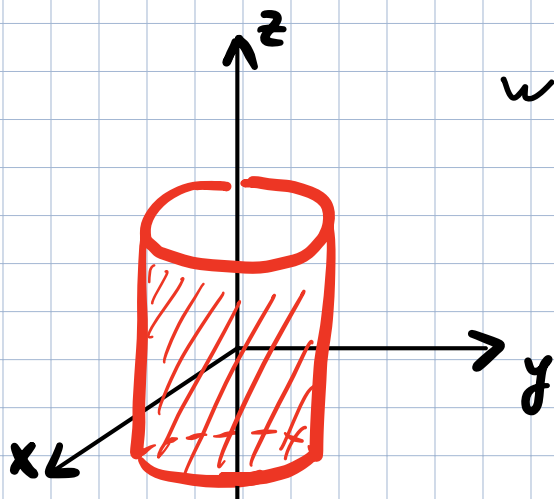
circle of radius 1
centered on z-axis

in the horizontal plane $z = 2$



(b)

$x^2 + y^2 = 1$ - cylinder of rad. 1
whose axis is z-axis



On $\mathbb{R}^2$: distance between
 $P(x_1, y_1)$ and $Q(x_2, y_2)$

is

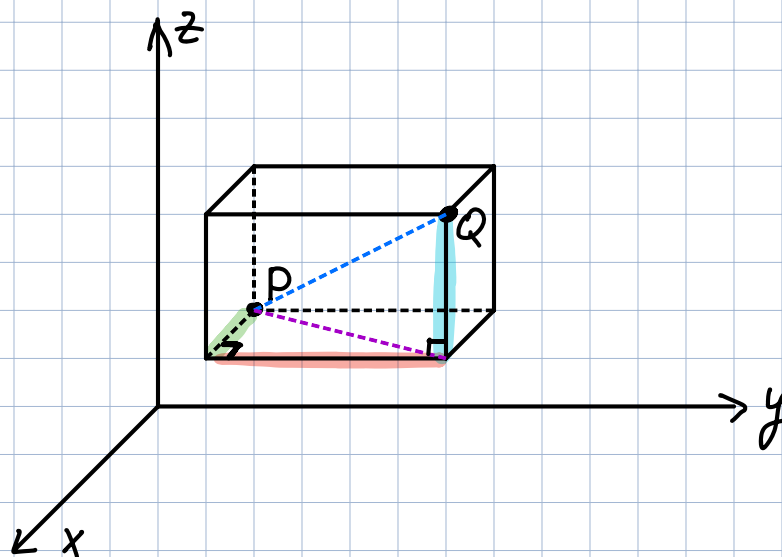
$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

In $\mathbb{R}^3$:

$P(x_1, y_1, z_1)$ $Q(x_2, y_2, z_2)$

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance:



Ex: $P(0, 1, -2)$ $Q(1, 3, 0)$

$$|PQ| = \sqrt{(1-0)^2 + (3-1)^2 + (0-(-2))^2} = \sqrt{1+4+4} = 3$$

A sphere of radius R with center $P(a, b, c)$ is given by equation

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = R^2$$

Rmk: In case $P=O$: $x^2 + y^2 + z^2 = R^2$

Ex: Show that the equation

$$x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$$

defines a sphere; find its radius and center.

$$\underline{x^2 + 4x} + \underline{y^2 - 6y} + \underline{z^2 + 2z} = -6$$

$$x^2 + 2 \cdot 2x + 2^2 \quad y^2 - 2 \cdot 3y + 3^2 \quad z^2 + 2z + 1^2$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = 4 + 9 + 1 - 6$$

$$(x - (-2))^2 + (y - 3)^2 + (z - (-1))^2 = 8$$

$$P(-2, 3, -1)$$

$$R = \sqrt{8} = 2\sqrt{2}$$